DATASHEET

HALCON Calibration Plate, Aluminum Composite

HALCON calibration plate with rectangularly arranged marks, the layout that HALCON's `gen_caltab` operator describes: black circular marks of one size in a regular grid on a white field, inside a black frame with a triangular mark in one corner. The frame lets `find_caltab` locate the plate and the triangle fixes its orientation. For use with `find_calib_object`, `find_caltab` and `calibrate_cameras`.

1 Ordering

Part number	Size (W × H)
CT-HAL-P-200X150-STD	200 × 150 mm
CT-HAL-P-300X200-STD	300 × 200 mm
CT-HAL-P-400X300-STD	400 × 300 mm
CT-HAL-P-600X400-STD	600 × 400 mm
CT-HAL-P-800X600-STD	800 × 600 mm

The number of marks, the mark distance and the mark diameter are fixed with the order: send your existing `.descr` file, or the `gen_caltab` parameters (marks in x and y, mark distance, diameter ratio), and the plate is made to them. A `.descr` description file in `gen_caltab` format is supplied with every plate. Plates with hexagonally arranged marks (`create_caltab`, `.cpd`): custom order (CUS).

Pattern area is centered with a white quiet zone ≥ 1 mark pitch on all sides; the edge strip with part number, serial and 100 mm scale bar sits outside the quiet zone.

2 Key specifications

ACCURACY	SUBSTRATE	FLATNESS
±0.3 mm + 0.2 mm/m (any mark center to nominal grid, at 20 °C)	Aluminum / LDPE / aluminum composite, 6 mm	≤ 0.5 mm per meter, board supported on three points

Unless stated otherwise, dimensions are nominal at 20 °C.

1 Specifications · geometry

FEATURE	SPECIFICATION
Feature position tolerance	±0.3 mm + 0.2 mm/m (any mark center to nominal grid, at 20 °C)
Mark pitch, mean	±0.05 mm
Circularity	≤ 0.05 mm radial deviation
Edge radius	R6 mm

2 Specifications · physical

FEATURE	SPECIFICATION
Substrate	Aluminum / LDPE / aluminum composite, 6 mm
Weight	6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg)
Pattern	Direct UV print, matte, black on white; optical density ≥ 1.6
Flatness	≤ 0.5 mm per meter, board supported on three points
Mounting	4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture CT-ACC-M-FIX14 clamps any size

3 Specifications · environment

FEATURE	SPECIFICATION
Operating temperature	−50 to +80 °C
Thermal expansion	≈ 24 µm/(m·°C)
Storage	−20 to +60 °C, dry
Outdoor lifetime	up to 3 years (UV-stable print)
Cleaning	water or ≤ 70 % isopropyl alcohol, soft cloth

4 Deliverables

INCLUDED		SERVICE	
Measurement	Every board is measured before it ships (dimensional, flatness, visual)	Lead time	Made to order · ships in 2–3 weeks
Files	`.halcon.descr`, `.opencv.json`, `.dxf`	Options	Logo print (-L) · hard case (CT-ACC-M-CSE400) · custom NRE
Documents	this datasheet, CT-UM-GEN user manual		

5 Compliance & customs

TOPIC	STATEMENT
RoHS	RoHS-compliant materials.
HS code	`9031 80` Not a toy. not for food contact.

6 Revision history

REV	DATE	CHANGE
A	`2026-10`	First issue, planning values

7 Disclaimer

Values are nominal unless stated as measured.